GRANT NEILSON

Computational Biologist with seven years of industry experience building scalable and robust analysis pipelines, developing internal Python & R packages and processing large-scale whole genome and single cell sequencing datasets (1TB+). Committed to writing production-quality code through rigorous testing, version control, and thorough documentation. Active open-source contributor, passionate about developing reliable scientific software and advancing reproducible research through community-driven collaboration.

View this CV online

INDUSTRY EXPERIENCE

2026 | 2023

Senior Computational Biologist

CoSyne Therapeutics — London, UK

- Led analysis of large-scale genomic datasets to support biomarker discovery and target identification, translating complex results into clear, decision-ready outputs for cross-functional teams. Culminated in the successful partnership with a publicly traded biotech.
- Co-Developed and maintained internal & open source R and Python packages. For statistical modelling, data processing and visualisation. Bixverse, is a Rust based R package enabling the analysis of 3 million single cells on a standard laptop. Cosyne-cnvutils is a python package used for the analysis of genomics data.
- Developed internal container system using Docker and ECR, to enable reproducible research. This system used CI/CD to build images robustly and securely.
- Recognized as a named contributor in version 3.4.1 of the NF-core Sarek pipeline. Optimised the running of the pipeline, achieving a 50% cost reduction and 30% decrease in computational time per sample.
- Co-developed an internal MCP server exposing CoSyne's omics lakehouse and public biomedical APIs to LLM agents. Authored the DuckLake SQL catalog and query layer: a curated, allowlisted table catalog encoding row grain, join keys and data caveats, with server-side SELECT-only validation and row caps forming the safety boundary for model-generated SQL. Connected to Claude, enabling agents to reason across internal and public data and generate testable hypotheses.
- Led ingestion for the company DuckLake lakehouse, a Postgres catalog over Parquet on S3 queried with DuckDB, landing 70+ schema-validated tables spanning internal RNA-seq, WGS, methylation, CRISPR and dose-response data alongside OpenTargets, DepMap, CCLE/PRISM and Orphanet. Built a polars and pandera validation layer enforcing a declared schema on every write, shipped via Docker/ECR images and GitLab and GitHub CI.

CONTACT

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GitHub
LinkedIn

LANGUAGES

R
Python
Bash
Nextflow
SQL

TECHNOLOGIES

GitLab/GitHub
Docker
AWS
Azure
DuckDB
MCP/LLM Agents

OPEN SOURCE & PUBLIC PROJECTS

bixverse
manifoldsR
cosyne-cnvutils
Sarek

*Made with the R package **pagedown**.*

The source code is available on GitHub.

Last updated October 2026.

2023 | 2020

Computational Biologist

Fios Genomics — Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

2020 | 2019

Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School — Exeter, UK

- Developed novel bioinformatics pipelines for processing genomics data.
- Gained an understanding of regression analysis, how to account for covariates, feature selection and optimal model fitting.

EDUCATION

2019 | 2015

MSci Biology

University of Southampton — Southampton, UK

- Grade : First Class Honours
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015 | 2013

School

Merchiston Castle School — Edinburgh, UK

- A Levels: Biology, Chemistry, Maths

SELECTED PUBLICATIONS

2025

TP53 and RB1 are predictive genetic biomarkers for sensitivity to cytarabine in gliomas

BioRxiv

- L. Laaniste, J. Luginbuehl, M. Trinh, R. Watkins, G. Lueg, **G. Neilson**, et al.
- doi: https://doi.org/10.1101/2025.11.20.688938